Homework Assignment

Best Square Approximation of Continuous Functions

Course Topic: Function Approximation and Least Squares Methods

Instructions. Answer all problems clearly.

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Problem 1. Gram Matrix Formulation

Let

$$V_n = \text{span}\{\phi_0, \phi_1, \ldots, \phi_n\}.$$

Assume the functions $\phi_0, \phi_1, \ldots, \phi_n$ are linearly independent. We seek

$$p_n(x) = \sum_{j=0}^{n} c_j \phi_j(x).$$

1. Derive the normal equations for the best square approximation of f from V_n.
2. Define the Gram matrix G and the right-hand side vector b.
3. Show that the normal equations can be written as

$$Gc = b.$$

4. Prove that G is symmetric positive definite.
5. Explain why the positive definiteness of G implies uniqueness of the best approximation.

Problem 2. Best Linear Approximation on $[0, 1]$

Find the best square approximation of

$$f(x) = x^2$$

on $[0, 1]$ by a linear polynomial

$$p(x) = c_0 + c_1 x.$$

1. Set up the least-squares error functional

$$E(c_0, c_1) = \int_0^1 \left(x^2 - c_0 - c_1 x\right)^2 \, dx.$$

2. Derive the normal equations.
3. Solve for c_0 and c_1.
4. Compute the squared error

$$\|x^2 - p(x)\|_2^2.$$

Problem 3. Orthonormal Basis Formula

Suppose

$$\{\psi_0, \psi_1, \ldots, \psi_n\}$$

is an orthonormal basis for V_n, meaning

$$\langle \psi_i, \psi_j \rangle = \delta_{ij}.$$

1. Show that the best square approximation to f from V_n is

$$p_n^*(x) = \sum_{k=0}^{n} \langle f, \psi_k \rangle \psi_k(x).$$

2. Derive the error identity

$$\|f - p_n^*\|_2^2 = \|f\|_2^2 - \sum_{k=0}^{n} |\langle f, \psi_k \rangle|^2.$$

3. Explain why adding more orthonormal basis functions cannot increase the best approximation error.

Problem 4. Legendre Polynomial Approximation

On $[-1, 1]$, the first three Legendre polynomials are

$$P_0(x) = 1, \qquad P_1(x) = x, \qquad P_2(x) = \frac{1}{2}(3x^2 - 1).$$

They satisfy

$$\int_{-1}^{1} P_i(x) P_j(x)\, dx = 0 \qquad \text{if } i \neq j,$$

and

$$\int_{-1}^{1} P_j(x)^2\, dx = \frac{2}{2j+1}.$$

Let

$$f(x) = x^3.$$

Find the best square approximation to f from

$$\text{span}\{P_0, P_1, P_2\}.$$

1. Write the approximation in the form

$$p_2(x) = a_0 P_0(x) + a_1 P_1(x) + a_2 P_2(x).$$

2. Use the formula

$$a_j = \frac{\langle f, P_j \rangle}{\langle P_j, P_j \rangle}$$

to compute a_0, a_1, a_2.

3. Write the final polynomial in both Legendre form and monomial form.

4. Explain why the coefficient of P_2 is zero or nonzero.

Problem 5. Weighted Best Square Approximation

Let $w(x) > 0$ be a continuous weight function on $[a, b]$. Define

$$\langle f, g \rangle_w = \int_a^b f(x)g(x)w(x)\,dx.$$

1. Define the weighted best square approximation to f from V_n.
2. Derive the weighted orthogonality condition

$$\langle f - p^*, q \rangle_w = 0 \qquad \text{for all } q \in V_n.$$

3. Derive the weighted Gram matrix system.
4. Explain how the weight function $w(x)$ affects the approximation.
5. Give an example of a situation where a weighted approximation may be more useful than an unweighted approximation.

Problem 6. Function Fitting with a Quadratic Polynomial

Let

$$f(x) = \sin(\pi x), \qquad -1 \le x \le 1.$$

Find the best square approximation of f from

$$P_2 = \text{span}\{1, x, x^2\}.$$

1. Set up the Gram matrix system using the monomial basis $\{1, x, x^2\}$.
2. Use symmetry to simplify the system as much as possible.
3. Determine the coefficients of the best quadratic approximation.
4. Explain why some coefficients must vanish due to symmetry.
5. Compare the result with the Taylor polynomial of $\sin(\pi x)$ at $x = 0$.